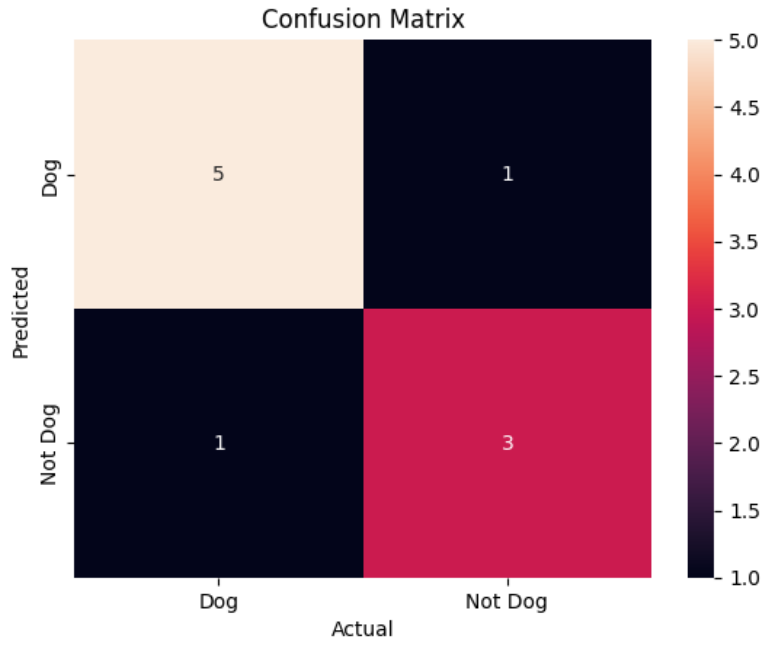
Experiment 1:

Aim: To demonstrate a confusion matrix using Python.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
actual = ['Dog','Dog','Dog','Not Dog','Dog',
          'Not Dog','Dog','Dog','Not Dog','Not Dog']
predicted = ['Dog','Not Dog','Dog','Not Dog','Dog',
             'Dog','Dog','Dog','Not Dog','Not Dog']
cm = confusion_matrix(actual, predicted)
sns.heatmap(cm, annot=True, fmt='d',
            xticklabels=['Dog','Not Dog'],
            yticklabels=['Dog','Not Dog'])
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Confusion Matrix")
plt.show()
```

Output:



Result: Thus, the confusion matrix was successfully demonstrated using Python.